Worksheet Part W5: the $p = 5$ table

accompanying *Mild p-class tower groups of imaginary quadratic fields*

References of the form Definition 3.17 or Part W7 refer to the statements of the paper and to the parts of this repository.

Part W5: The $p = 5$ table

The 204 imaginary quadratic fields with 5-class rank three and $|D_K| < 2^{30}$, one line per field. The columns record the class group, the number of $\mathbb{F}_5$-rational points of the norm-degeneracy scheme Σ_D (Definition 3.17 of the paper) at which $\operatorname{rk} D_x = 1$, the number of these which are transverse, and the least degree of an extension of $\mathbb{F}_5$ over which a transverse point exists. The defining polynomials of the extensions used for the fields without a rational transverse point, the coordinates of all transverse points, and the corresponding transversality matrices are recorded for every field in `records/p5/transverse-rank-one/` (`report.txt` human-readable, `certificates.json` machine-readable); the finite-field calculations underlying the table are re-run deterministically by `tools/transverse_rank_one.py`. For the field $D_K = -781922404$, marked *, the scheme Σ_D is a single non-reduced closed point of degree two, so no transverse point exists over any extension; the field is mild by a word count after a change of variables over $\mathbb{F}_{25}$, recorded in Part W7.

	D_K	$\mathrm{Cl}(K)$	rank-one over $\mathbb{F}_5$	transverse over $\mathbb{F}_5$	least transverse degree
1	-11203620	$[10, 10, 10]$	2	1	1
2	-18397407	$[40, 10, 5]$	3	3	1
3	-27960639	$[40, 10, 10]$	2	2	1
4	-35663739	$[30, 10, 5]$	0	0	6
5	-51213139	$[75, 5, 5]$	1	0	4
6	-54319112	$[60, 10, 5]$	4	2	1
7	-61040707	$[20, 10, 5]$	2	1	1
8	-65818135	$[30, 10, 10]$	3	2	1
9	-75949255	$[40, 20, 5]$	2	2	1
10	-77778287	$[120, 10, 5]$	3	3	1
11	-89017304	$[170, 10, 5]$	2	2	1
12	-89218664	$[310, 5, 5]$	1	1	1
13	-90903207	$[110, 10, 5]$	2	2	1
14	-93121640	$[90, 10, 10]$	1	1	1
15	-104545864	$[60, 10, 5]$	1	1	1
16	-106660295	$[110, 10, 10]$	2	2	1
17	-109909943	$[100, 10, 10]$	1	1	1
18	-123482119	$[270, 5, 5]$	4	4	1
19	-123779560	$[10, 10, 10, 2]$	2	1	1
20	-126740891	$[195, 5, 5]$	0	0	6
21	-129063459	$[60, 10, 5]$	1	1	1
22	-130370807	$[100, 10, 10]$	0	0	2
23	-134005207	$[100, 10, 5]$	3	2	1
24	-145367147	$[125, 5, 5]$	2	2	1
25	-150535579	$[140, 5, 5]$	2	2	1
26	-150679519	$[150, 10, 5]$	0	0	6

	D_K	$\mathrm{Cl}(K)$	rank-one over $\mathbb{F}_5$	transverse over $\mathbb{F}_5$	least transverse degree
27	-150752699	$[160, 5, 5]$	2	1	1
28	-156145931	$[165, 5, 5]$	3	3	1
29	-164995035	$[60, 10, 5]$	1	0	2
30	-165570488	$[60, 20, 5]$	1	1	1
31	-167094823	$[60, 10, 10]$	1	1	1
32	-176386616	$[180, 10, 5]$	2	1	1
33	-181738571	$[100, 10, 5]$	2	1	1
34	-182240771	$[225, 5, 5]$	3	3	1
35	-186243704	$[40, 20, 10]$	0	0	3
36	-187160831	$[60, 20, 10]$	2	1	1
37	-190572839	$[630, 5, 5]$	2	2	1
38	-194080808	$[70, 10, 10]$	1	1	1
39	-199674799	$[340, 5, 5]$	1	1	1
40	-202331347	$[30, 10, 5]$	1	1	1
41	-203145767	$[60, 10, 10, 2]$	2	1	1
42	-203437103	$[210, 10, 5]$	3	2	1
43	-204909944	$[80, 10, 10]$	1	1	1
44	-205996583	$[100, 10, 10]$	1	1	1
45	-207666763	$[50, 10, 5]$	1	1	1
46	-209149784	$[180, 10, 5]$	3	2	1
47	-212229543	$[60, 10, 10]$	1	0	3
48	-216770467	$[95, 5, 5]$	1	1	1
49	-225826216	$[20, 10, 10, 2]$	1	1	1
50	-226373428	$[30, 10, 10]$	3	3	1
51	-231112315	$[20, 10, 10]$	2	2	1
52	-239452895	$[160, 10, 10]$	1	1	1
53	-242963687	$[370, 5, 5]$	0	0	6
54	-246238555	$[70, 10, 5]$	1	1	1
55	-247127879	$[160, 10, 10]$	1	1	1
56	-248046051	$[60, 10, 5]$	1	1	1
57	-249407767	$[60, 10, 10]$	0	0	6
58	-252735992	$[280, 5, 5]$	5	4	1
59	-253425979	$[135, 5, 5]$	0	0	3
60	-256232447	$[420, 5, 5]$	1	1	1
61	-261286872	$[10, 10, 10, 2, 2]$	1	1	1
62	-272394484	$[150, 10, 5]$	0	0	6
63	-280557591	$[150, 10, 10]$	2	1	1
64	-282198803	$[120, 10, 5]$	1	1	1
65	-285889891	$[60, 10, 5]$	2	1	1
66	-294897647	$[100, 10, 10]$	0	0	6
67	-300240404	$[250, 10, 5]$	1	1	1
68	-308728199	$[820, 5, 5]$	0	0	2
69	-311069163	$[60, 10, 5]$	1	1	1
70	-312163691	$[130, 10, 5]$	1	1	1
71	-313065188	$[140, 10, 5]$	1	1	1
72	-330727336	$[150, 5, 5]$	0	0	6
73	-334692215	$[80, 10, 10, 2]$	0	0	2
74	-342156463	$[200, 10, 5]$	5	4	1
75	-343576036	$[220, 5, 5]$	0	0	2
76	-344438804	$[300, 10, 5]$	0	0	2
77	-345369167	$[160, 10, 10]$	3	2	1

	D_K	$\mathrm{Cl}(K)$	rank-one over $\mathbb{F}_5$	transverse over $\mathbb{F}_5$	least transverse degree
78	-348003156	$[70, 10, 10]$	2	1	1
79	-351044647	$[280, 5, 5]$	2	2	1
80	-360803647	$[310, 5, 5]$	1	1	1
81	-364780660	$[70, 10, 10]$	1	1	1
82	-368523908	$[20, 20, 10, 2]$	0	0	2
83	-416349256	$[40, 10, 10]$	2	1	1
84	-417286132	$[110, 10, 5]$	1	1	1
85	-429818367	$[120, 10, 10]$	1	1	1
86	-433597399	$[240, 10, 5]$	1	1	1
87	-434084164	$[300, 5, 5]$	0	0	2
88	-439073371	$[70, 10, 5]$	2	1	1
89	-443585299	$[100, 10, 5]$	2	1	1
90	-444408443	$[120, 10, 5]$	1	1	1
91	-448376516	$[110, 10, 10]$	3	2	1
92	-467347512	$[20, 10, 10, 2]$	2	2	1
93	-469835647	$[80, 20, 5]$	1	1	1
94	-480094015	$[350, 5, 5]$	2	1	1
95	-480642151	$[270, 10, 5]$	1	0	4
96	-482644063	$[100, 10, 10]$	1	1	1
97	-498607924	$[160, 10, 5]$	1	1	1
98	-501070231	$[775, 5, 5]$	3	1	1
99	-506335479	$[190, 10, 10]$	1	1	1
100	-508163803	$[30, 10, 10]$	0	0	2
101	-509504379	$[140, 10, 5]$	0	0	6
102	-512578628	$[80, 10, 10]$	0	0	2
103	-516168299	$[450, 5, 5]$	0	0	2
104	-520957412	$[230, 10, 5]$	2	1	1
105	-522401819	$[320, 5, 5]$	2	2	1
106	-527734248	$[80, 10, 10]$	1	1	1
107	-541579031	$[280, 10, 10]$	1	1	1
108	-545873304	$[70, 10, 10]$	2	1	1
109	-558181271	$[1095, 5, 5]$	1	1	1
110	-568232120	$[50, 10, 10, 2]$	3	2	1
111	-573379016	$[60, 10, 10, 2]$	2	1	1
112	-578286527	$[110, 10, 10, 2]$	2	1	1
113	-583175288	$[50, 10, 10, 2]$	0	0	2
114	-587516655	$[40, 10, 10, 2, 2]$	2	2	1
115	-587585540	$[60, 10, 10, 2]$	1	1	1
116	-600487131	$[80, 10, 10]$	1	1	1
117	-611642824	$[50, 10, 10]$	1	1	1
118	-614889363	$[20, 20, 10]$	2	2	1
119	-617556791	$[270, 10, 10]$	2	2	1
120	-618229927	$[360, 5, 5]$	1	1	1
121	-638094767	$[730, 5, 5]$	1	1	1
122	-648249064	$[70, 10, 10]$	3	3	1
123	-648716003	$[300, 5, 5]$	0	0	2
124	-652187816	$[130, 10, 10]$	2	2	1
125	-654932024	$[330, 10, 5]$	0	0	6
126	-659084579	$[570, 5, 5]$	0	0	2
127	-663884564	$[140, 10, 10]$	1	1	1
128	-671772331	$[320, 5, 5]$	2	2	1

	D_K	Cl(K)	rank-one over $\mathbb{F}_5$	transverse over $\mathbb{F}_5$	least transverse degree
129	-672002387	$[130, 10, 5]$	1	1	1
130	-684312527	$[600, 10, 5]$	1	0	4
131	-690429160	$[20, 10, 10, 2]$	5	4	1
132	-694030820	$[130, 10, 10]$	2	2	1
133	-694740327	$[240, 10, 5]$	0	0	6
134	-703985192	$[390, 5, 5]$	1	1	1
135	-716214548	$[130, 10, 10]$	0	0	6
136	-716545112	$[310, 10, 5]$	2	2	1
137	-717538411	$[215, 5, 5]$	2	1	1
138	-719003311	$[340, 10, 5]$	1	1	1
139	-720658367	$[390, 10, 5]$	3	3	1
140	-729506291	$[520, 5, 5]$	2	1	1
141	-730579195	$[40, 20, 5]$	1	1	1
142	-736568551	$[705, 5, 5]$	4	4	1
143	-747498712	$[100, 10, 5]$	0	0	2
144	-747906091	$[210, 5, 5]$	4	4	1
145	-752704467	$[130, 5, 5]$	4	2	1
146	-753660888	$[60, 10, 10]$	1	1	1
147	-758811723	$[110, 10, 5]$	1	1	1
148	-770289259	$[275, 5, 5]$	1	1	1
149	-781922404	$[120, 15, 5]$	0	0	—*
150	-792236011	$[130, 10, 5]$	2	2	1
151	-818819903	$[320, 10, 5]$	2	2	1
152	-822797303	$[480, 10, 5]$	2	2	1
153	-823763252	$[200, 10, 5]$	0	0	6
154	-829075271	$[1490, 5, 5]$	3	2	1
155	-832956936	$[50, 10, 10, 2]$	1	1	1
156	-834867768	$[30, 10, 10, 2]$	1	1	1
157	-835199035	$[50, 10, 10]$	3	2	1
158	-837148756	$[150, 10, 5]$	0	0	3
159	-839189588	$[160, 10, 10]$	3	3	1
160	-847946303	$[1020, 5, 5]$	0	0	3
161	-848798407	$[375, 5, 5]$	2	2	1
162	-856365643	$[170, 5, 5]$	3	3	1
163	-859104103	$[230, 10, 5]$	0	0	3
164	-860862356	$[710, 5, 5]$	3	2	1
165	-860938607	$[390, 10, 5]$	0	0	2
166	-863732452	$[90, 10, 5]$	2	1	1
167	-864855431	$[60, 60, 10]$	1	1	1
168	-865276015	$[280, 10, 5]$	0	0	3
169	-865288212	$[60, 10, 10, 2]$	3	3	1
170	-865931992	$[60, 10, 10]$	1	1	1
171	-873927415	$[240, 10, 5]$	1	1	1
172	-876422059	$[265, 5, 5]$	1	0	4
173	-880112408	$[270, 10, 5]$	1	1	1
174	-880342436	$[660, 5, 5]$	1	1	1
175	-881852344	$[80, 20, 5]$	4	4	1
176	-882321235	$[130, 10, 5]$	0	0	6
177	-884383779	$[180, 10, 5]$	2	2	1
178	-890032871	$[750, 10, 5]$	2	1	1
179	-912814943	$[600, 10, 5]$	2	2	1

	D_K	$\mathrm{Cl}(K)$	rank-one over $\mathbb{F}_5$	transverse over $\mathbb{F}_5$	least transverse degree
180	-930871220	$[160, 10, 10]$	0	0	6
181	-937521204	$[60, 20, 10]$	2	2	1
182	-941867380	$[180, 10, 5]$	1	1	1
183	-954227011	$[270, 5, 5]$	0	0	6
184	-964632455	$[1180, 5, 5]$	2	2	1
185	-967077539	$[450, 5, 5]$	1	1	1
186	-975828939	$[270, 5, 5]$	3	2	1
187	-976016443	$[210, 5, 5]$	1	1	1
188	-979106379	$[410, 5, 5]$	0	0	6
189	-996554680	$[60, 10, 10]$	0	0	2
190	-1007496479	$[920, 10, 5]$	1	1	1
191	-1008688623	$[850, 5, 5]$	1	0	4
192	-1011574052	$[60, 20, 10]$	1	0	3
193	-1014273539	$[360, 5, 5]$	0	0	6
194	-1016848324	$[300, 5, 5]$	4	4	1
195	-1018718571	$[180, 10, 5]$	1	1	1
196	-1019215031	$[450, 15, 5]$	0	0	6
197	-1030705032	$[40, 10, 10, 2]$	1	1	1
198	-1040930248	$[120, 10, 5]$	2	2	1
199	-1041244847	$[470, 10, 5]$	0	0	2
200	-1043658971	$[550, 5, 5]$	0	0	2
201	-1047634431	$[1030, 5, 5]$	1	1	1
202	-1064940171	$[180, 10, 5]$	2	2	1
203	-1066461896	$[330, 10, 5]$	0	0	6
204	-1069759951	$[360, 10, 5]$	2	2	1